

SG DATA PACK

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01	02	03
ID		RecrawlCandidate

□ □ □ □ · · · AI

□ □ □ □ codex/agent-task-experiments d76bb32

SG Data PackJavaScriptHTMLJSONData Pack

“JSON”

SG Data Pack	
	slug ID + aliases + sameAs
	stageoverlay
	E1-E16 fail loudly + Review Decisions
JSON	

1. Data Pack
2. recrawlCandidate
3. AgentAI

“ ” “ ”

01

HTML

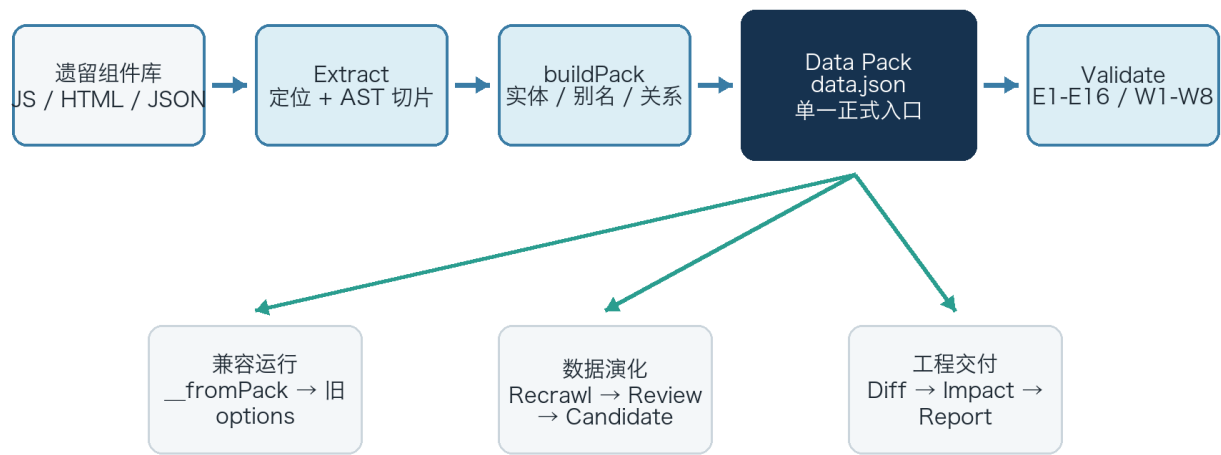
- IIFE UMD options fallback
-
- HTML <script type="application/json">
-
-
-

engine fallback

API
recrawl
AI

02

SG Data Pack: 从遗留数据到可信变更链



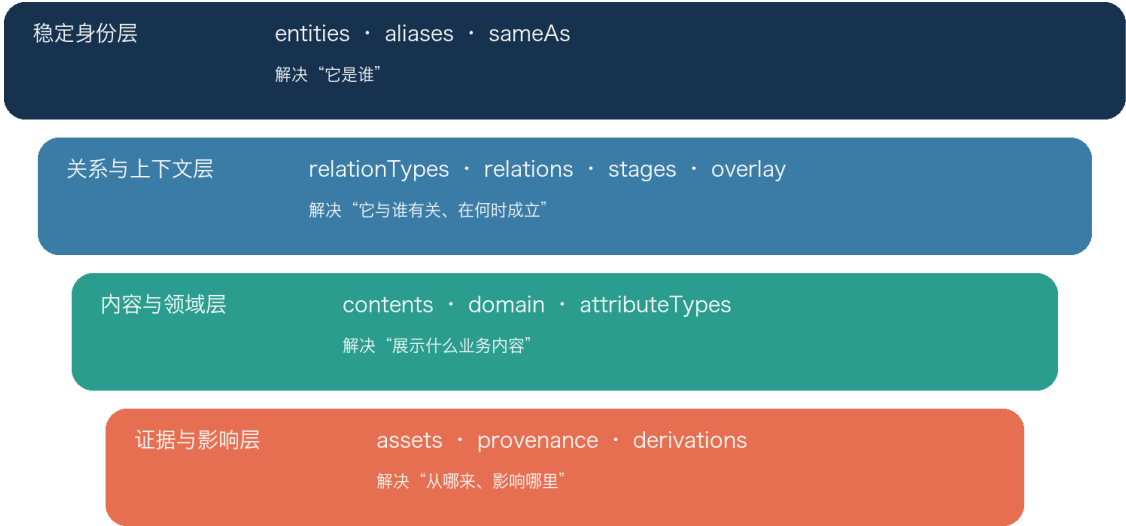
设计重点：规范化不是终点；可验证、可审核、可解释的变更链才是产品核心。

HTML JSON script		Data Pack	
Baseline + crawl records + Decisions		Candidate + Audit	
Validation + Rules + Diff + Evidence		RunReport	CI

- Data Pack renderer __fromPack options
- Fail closed
- SHA-256 “ ”
-
- domain

03 Data Pack

Data Pack 的分层结构



通用核心保持严格；组件库差异进入 domain，并由库级 schema / rules 补充。

entities	ID
aliases	canonical ID
relationTypes / relations	master edge
stages / overlay	
contents / domain	
assets	hash
provenance	
derivations	

04

```
entities.chengbing = { kind: "person", name: " " }
aliases[" "] = "chengbing"
relations[0] = { a: "chengbing", b: "wangdayong", type: "rival" }
```

ID

+

```
relations: [
  { id: "a-b-advisor", a: "a", b: "b", type: "advisor", scope: ["stage-1"] }
]

stages: [
  { key: "stage-1", entities: ["a", "b"],
    relations: [{ a: "a", b: "b", id: "a-b-advisor" }],
    overlay: { a: { role: " " } } }
]
```

-
- scope stage ref
- overlay
- Diff

contextual alias	context	id
a → b		canonical ordering
relation identity	type/scope	provenance key
ID rename/migration		

05

__fromPack

Data Pack options Branch by Abstraction

renderer	Data Pack
JSON renderer	
Data Pack → __fromPack → options	deep equivalence

- 4.
- 5. Acorn parseExpressionAt
- 6. buildPack entities aliases relations domain
- 7. __fromPack defaults deep-equal
- 8. data.json data.js data.schema.json

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extraction config	literal DOM mount

- Data Pack engine fallback
- ctx
- new Function require config require engine
- CI compile --check / compare-existing fallback

06 Fail Loudly

JSON Schema	schemaVersion	meta	relation item
SGDataLoader	stage	provenance	derivation
	path		

E1-E16

	schemaVersion	meta.id	meta.title
	slug	alias target	ID
	scope	stage edge	resolution
		hash	
	sameAs	record/field	provenance
		sourceUrl	confidence
	derivation kind	source	alsoTouches consumers affects

mount	CI		
JSON Schema		loader	
Warnings		strict/non-strict	
Node	UMD	validate	500+

07

provenance

provenance

```
provenance.entities.alice = {
  origin: "crawl:reviewed",
  sourceUrl: "https://example.test/source",
  confidence: 0.95,
  fieldOrigins: { occupation: { origin: "crawl:reviewed", confidence: 0.95 } }
}
```

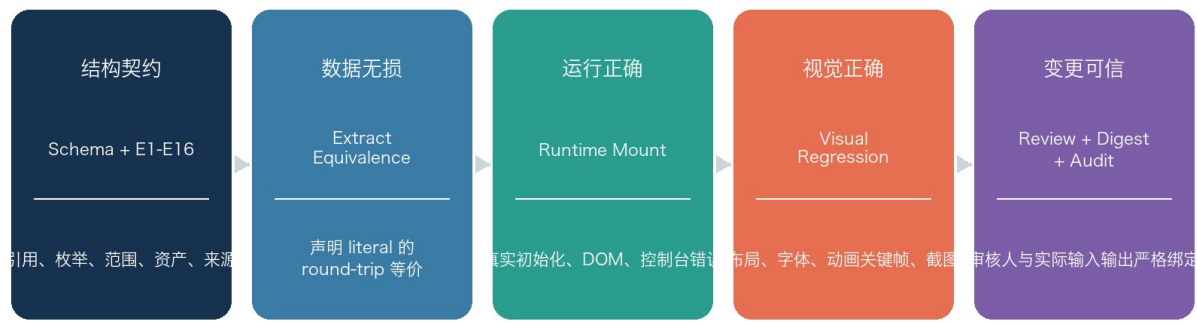
derivations

```
derivations.carouselTrack = {
  kind: "repeat",
  source: "domain.carouselTrack.entityIds",
  alsoTouches: ["entities.*"],
  consumers: ["components/carousel-item.js"],
  affects: [".carousel-track"],
  note: " "
}
```

provenance			
confidence + threshold		review	
derivation source	Diff		
affects / consumers		validator	selector

08

证据分层：每一层只证明它真正覆盖的范围



关键原则：不能用低层证据替代高层证据。JSON deepEqual 通过，不等于页面视觉已经验证。

“JSON” “Schema” “deepEqual” “SG Data Pack” assurances

v1.3 Pilot	188 / 188			0 failed	5.88
	0 error / 0 warning			ID-based	Collection
	entity + asset + derivation			work-beta.cover	
	byte-exact			HTML	
	70.01%	66.56%	71.06%	6333	vendored Acorn
□□□□□□□					
library	78%–100%		runtime/visual	Pilot	

09 Recrawl Candidate

Data Pack

```
records.json
  ↓ cross-check
agree / gap / conflict / unsupported / identity miss
  ↓ explicit decisions
apply / keep / map-alias / reject
  ↓ candidate validation
candidate.json + candidate-audit.json
```

- Review Report baseline/records bytes digest
- Candidate cross-check
- review item decision
- last-write-wins
-
- Audit before after reviewer confidence diff derivation impact

alias	alias
field-level provenance	nested value
apply / keep / map-alias / reject	relations stages contents domain assets derivations

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Candidate	Data Pack

10 Library Evolution Report

/

- JSON Markdown RunReport
- Outcome ready blocked review-required input-error
- Maturity errors extraction runtime visual
- findings assurances impacts

collectReport digest review/audit diff validate()

PackCollector RulesCollector ExtractionCollector ReviewCollector CandidateAuditCollector DiffCollector

collector findings / assurances / risks / artifacts

11 Agent “AI”

Data Pack AI Agent patch
AgentTaskManifest → TaskRun → ExperimentReport

Manifest	instructions evidence	source tree	input bytes	file policy	grader
Agent	patch				
Patch apply	apply	allow/deny			
Grading	digest-verified staging	grader contract/rules/command/runtime/visual checks			
Experiment	Wilson 95%				

“ ”

- OS sandbox
-
-
-
- Agent grader secrecy

			Agent benchmark		Data Pack
			tracked files	263	research/agent-eval/results
			worker/subprocess		
			patch parser	path policy	tree snapshot grader experiment

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	ID	alias	
Reference, never duplicate		stage	overlay
	__fromPack		DOM/
Fail loudly		mount/CI	
	provenance + derivations + diff		
	crawl	Decisions	Candidate
	JSON hash	ID	CI
	Node 18+	loader	Agent skill

9.
10.
11. AI patch policy grader evidence

13

	Schema loader types docs	Candidate	
	validate collectReport	runTask	
	pack	embedded fallback	
	new Function + require config/engine		
	/		
domain			
DSL	dot path + wildcard		key
Candidate			
Agent	CLI		
	glob	CLI diff	patch

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“ ”

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P0

SGDataLoader.validate		meta/entities/relations/stages/assets/provenance/derivations		
Contract Registry		Schema/TS/docs		registry
collectors	collectReport	evidence		
Agent Eval	packages/agent-eval	CLI	Data Pack	core
			worker/process isolation	sandbox

P1

- context alias resolution
- entity ID rename/migration relations stages sameAs provenance
- Candidate relation/stage precondition + allow-listed diff
- derivation/attributeSources JSON Pointer
- hash SHA-256 asset source paths

P2

- package.json bin engines
- macOS/Windows CI vendor
- Agent trial release artifact digest
- Data Pack merge conflict

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AI

			/		
Core Contract			validate compile loader types		
Migration Kit			extract templatize engine integration		
Evolution Workflow		/	recrawl-skeleton candidate diff report		
Library Rules			rules domain schema DATA-GUIDE		
Agent Evaluation		AI	task grade evidence experiment		

“ data.json”

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- AI

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SG Data Pack Data Pack Fail Loudly Review Candidate

Data Pack

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- Agent benchmark Data Pack

- Core
- Evolution
- Agent Eval AI

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SG Data Pack

APPENDIX

A

	/Users/tangyaoyue/DEV/sg-data-pack/README.md
Agent Skill	/Users/tangyaoyue/DEV/sg-data-pack/SKILL.md
Data Pack	/Users/tangyaoyue/DEV/sg-data-pack/references/data-pack-contract.md
	/Users/tangyaoyue/DEV/sg-data-pack/references/engine-integration.md
	/Users/tangyaoyue/DEV/sg-data-pack/references/extraction-config.md
	/Users/tangyaoyue/DEV/sg-data-pack/scripts/lib/sg-data-loader.js
	/Users/tangyaoyue/DEV/sg-data-pack/scripts/lib/sg-pack-extract-core.js
Recrawl Review	/Users/tangyaoyue/DEV/sg-data-pack/scripts/lib/sg-recrawl-review.js
Candidate	/Users/tangyaoyue/DEV/sg-data-pack/scripts/lib/sg-pack-candidate.js
	/Users/tangyaoyue/DEV/sg-data-pack/scripts/sg-pack-report.js
Agent Task Runner	/Users/tangyaoyue/DEV/sg-data-pack/scripts/lib/sg-task-runner.js
	/Users/tangyaoyue/DEV/sg-data-pack/tests/

B

9 / 10	overlay
9 / 10	
9 / 10	Fail closed digest binding evidence
7 / 10	domain
6.5 / 10	
6 / 10	OS sandbox
6.5 / 10	Agent Eval